stupidities, I brought their equations into precisely the needed form, implemented them in Mathematica, and wrote them up for Oliver, who incorporated them into his own computer code. At last his code could produce the quality images needed for the movie.

At Double Negative, Oliver's computer code was just the beginning. He handed it over to an artistic team led by Eugénie von Tunzelmann, who added an accretion disk (Chapter 9) and created the background galaxy with its stars and nebulae that Gargantua would lens. Her team then added the *Endurance* and Rangers and landers and the camera animation (its changing motion, direction, field of view, etc.), and molded the images into intensely compelling forms: the fabulous scenes that actually appear in the movie. For further discussion, see Chapter 9.

In the meantime, I puzzled over the high-resolution film clips that Oliver and Eugénie sent me, struggling to extract insights into why the images look like they look, and why the star fields stream as they stream. For me, those film clips are like experimental data: they reveal things I never could have figured out on my own, without those simulations—for example, the things I described in the previous section (Figures 8.5 and 8.6). We plan to publish one or more technical papers, describing the new things we learned.

Imaging a Gravitational Slingshot

Although Chris chose not to show any gravitational slingshots in *Interstellar*, I wondered what they would look like to Cooper as he piloted the Ranger toward Miller's planet. So I used my equations and Mathematica to simulate them and produce images. (My images have far lower resolution than Oliver's and Eugénie's due to my code's slowness.)